

Advanced Data Management

Information Integration Systems

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Outline of the course

1. Introduction to propositional logic
 2. Introduction to First-order Logic
 3. Relational Calculus
 - 4. Information Integration Systems (IIS)**
 5. Logical formalization of IISs
 6. Mapping between **Global Schema** e **Data Sources**
 7. Incomplete information databases
 8. Query answering over IISs
 - 9. Ontology-Based Data Management (OBDM)**
 10. Description Logic Ontologies
 11. Query answering in ontologies
 12. Query answering in OBDM
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Problem Definition

Information Integration

Information integration is the process of **reconciling** information coming from **multiple**, **autonomous**, and **heterogeneous** sources.

Information Management

What are the tasks of interest?

- **Query.**

- Answering user-specified queries over the reconciled view of the information at the sources.

- **Update, Delete, and Insert.**

- Updating, deleting, and inserting information using the reconciled view of the information at the sources.
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Physical vs Conceptual Heterogeneity

- **Physical Heterogeneity:** Information is represented using different formats. Datatypes and data models of the information sources may differ.
 - **Conceptual Heterogeneity:** The information sources describe the world from different perspectives.
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Physical Heterogeneity: Examples

- **Legacy databases**
 - Hierarchical model
 - Network model
- **Post-Relational**
 - Object Databases
 - Multi-value DBs
 - Graphs
- **SQL database.**
 - We can use SQL or a dialect thereof.
 - Relational Data Model.
- **NoSQL Databases**
 - Key-value stores
 - Big Tables
 - Documents (Json or the like)

Conceptual Heterogeneity: Example

Name	DOB	Residency
Amber A.	10/20/2030	New Amsterdam
Rose B.	10/20/1930	Constantinople

ID	Customer Name	Product
O001	Amber A.	P001
O002	Dylan B.	P001

Both databases are relational (so, no physical heterogeneity)

- What are the persons?
- What's their DOB?

Contexts of Application

Information Integration has mainly two contexts of application.

- **Intra-organizational.**
 - Different entities of an organization produce and consume information.
 - How can they communicate?
 - **Inter-organizational.**
 - Different organizations produce and consume information.
 - How can they communicate?
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Classifications

Information Integration Systems

An information Integration System is a piece of software that reconcile information coming from multiple sources.

We talk about **data integration tools** to refer to information integration systems that directly deal with data (in the sense of databases)

Classifications of Information Integration Systems

Information integration systems can be classified according to different characteristics.

- **Scope of the Integration:** What is the extent to which information should be integrated?
 - **System Specification:** How is the integration process specified?
 - **Information Replication and Timeliness:** What is the end result of the integration?
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Classifications

Scope of the integration

Scope of the Integration

According to the scope of the integration process, we classify information integration systems in three categories.

- **Task-Based Systems (one-shot):** Systems built to perform a single integration task.
 - **Service-Based Systems (query based):** Systems built to perform a specific task (query) over time.
 - **Domain Integration System (general purpose):** Systems built to perform user-specified tasks (queries).
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Task-Based Information Integration Systems

Task-Based systems perform a single specific task.

- Extract specific information from the sources at a specific point in time.

They are not meant for reuse:

- Code is written specifically for the task and may be discarded after use.

Possible applications:

- Data migration (loading pre-existing data into a new system)
 - Research-specific tasks (extracting business insight from data)
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Task-Based Systems: Example 1

Data Migration Task:

Data Sources: customer data from different branches of an acquiree company.

Task: Load data into the information system of the acquiring company.

Task-Based Systems: Example 2

Research-Specific Task:

Data Sources: Occupational data from the EU countries to date.

Task(s):

- Extract occupancy rate trend for the last century.
 - Extract agricultural workers trend over the last century.
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Service-Based Information Integration Systems

Service-Based systems perform a **specific task** over time.

- Extract specific information from the sources whenever is needed.

Meant for reuse:

- Service-Based systems may use user interfaces to facilitate interaction.
- Data may be a snapshot taken at a given point in time or refreshed at every query.

Limited Scope:

- While task may be repeated, the user have no freedom to choose the task.

Possible applications:

- Parametric query execution (content aggregators)
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Service-Based Systems: Example 1

Content Aggregator:

Data Sources: availability and cost of flights from different companies.

Task: Search for the cheapest flight for a given destination and date.

Service-Based Systems: Example 2

Content Aggregator:

Data Sources: prices of different goods from different sellers.

Task: Search for the cheapest product of a kind specified by the user.

Domain-Based Information Integration Systems

Domain-Based I.I.S. allow users to specify queries over the domain.

- Extract specific information from the sources whenever is needed.

Meant for reuse and broad scope:

- Domain-Based systems may use user interfaces to facilitate interaction.
- Data may be a snapshot taken at a given point in time or refreshed at every query.
- Different tasks (queries) can be performed over time.

Possible applications:

- Enterprise System Integration.
 - Auditing.
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Domain-Based Systems: Example

Auditing:

Data Sources: account data from multiple banks.

Task: Perform queries to track money laundering activities.

Domain-Based Information Integration

- **Domain-Based Information Integration** is the problem of providing a **unified view** of a collection of heterogeneous and autonomous sources.
- Essentially, our idea of information integration is domain-base integration.
- The **unified view** is achieved using a **global (or target) schema**, i.e., a formal definition of what the **result of the integration should look like**.
- Information at the sources is described by source schema(s), i.e., formal definitions how information looks like at the **sources**.

Domain-Based Information Integration

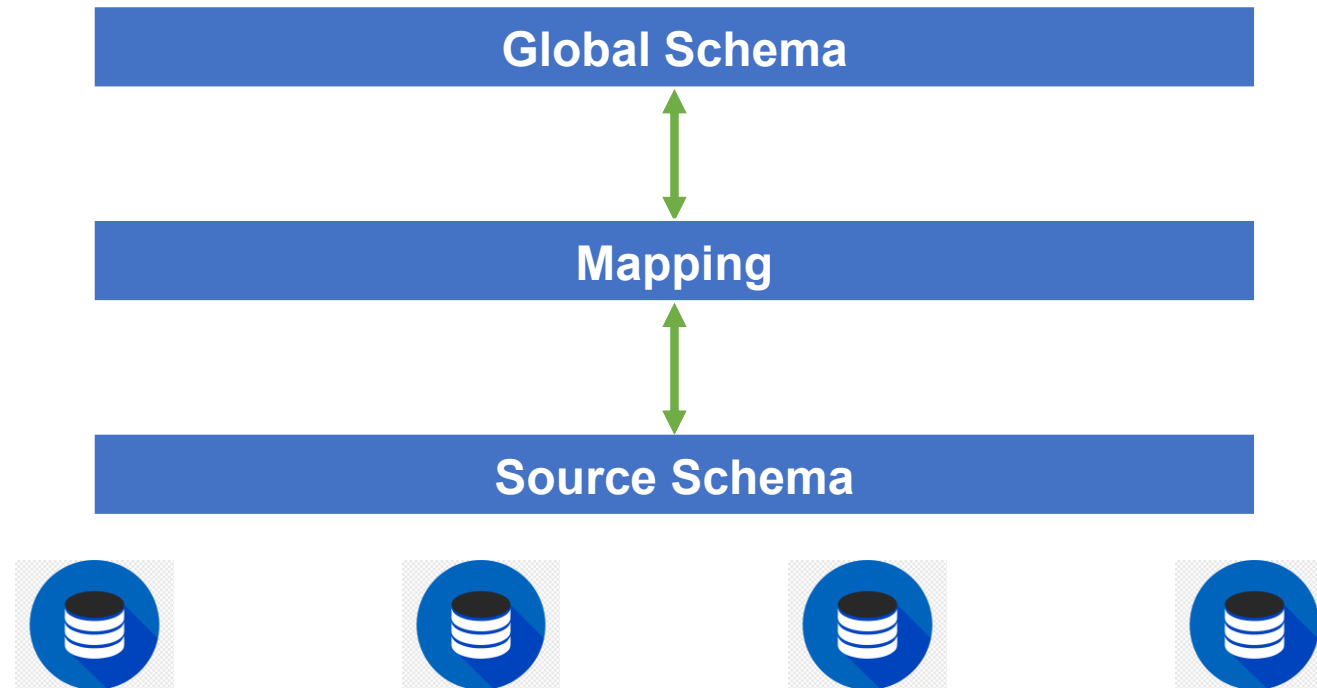
A **mapping layer** describes the **target data** (data in unified view) in terms of the **source data**.

Different approaches are possible

- **Ex 1:** Describing queries over the target schema in terms of queries over the sources.
- **Ex 2:** Sets of logical axioms
- **Ex 3:** Define data transformation processes.

From now on, we focus on Domain-Based Integration, so we will refer to Domain-Based Integration simply as Information Integration.

Information Integration Framework



Classifications

System Specification (Mapping Specification)

Mapping Specification

There are several ways to specify the **mapping** of an information integration system

Declarative

- The system is defined by specifying **what information** should be extracted
- Often using formal tools such as **first-order logic**.

Procedural (query-by-query)

- Mapping is defined by specifying the **process that should be used to perform the tasks**.
- Often using programming languages or procedural query languages (relational algebra)

Data Sources – Example

Name	DOB	Residency
Amber A.	10/20/2030	New Amsterdam
Rose B.	10/20/1930	Constantinople

ID	Customer Name	Product
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Global Schema: persons and an id code

Declarative Mapping Specification: Example

Source Schema: Customers(Name, DOB, Residency)
Purchases(Id, Customer_Name, Product)

Target Schema: Persons(Id, Name)

Declarative Mapping Definition (FOL):

- $\forall n, d, r. \text{Customers}(n, d, r) \rightarrow \exists i. \text{Persons}(i, n)$
- $\forall i_p, p, n. \text{Purchases}(i_p, n, p) \rightarrow \exists i. \text{Persons}(i, n)$

Procedural Mapping Specification: Example 1

Source Schema: Customers(Name, DOB, Residency)
Purchases(Id, Customer_Name, Product)

Target Schema: Persons(Id, Name)

Procedural Mapping Definition (Relational Algebra):

- Names: $Proj_1(Customers) \cup Proj_3(Purchases)$
- A mechanism to assign ids (more than relational algebra)

Procedural Mapping Specification: Example 2

Source Schema: Customers(Name, DOB, Residency)
Purchases(Id, Customer_Name, Product)

Target Schema: Persons(Id, Name)

Procedural Mapping Definition (Software Based):

- Software modules written to extract the data from the sources.

Implementing the Mapping

When mappings are defined **declaratively**, we need an implementation

Ad-hoc Software

- Build a system from scratch that implements the mapping
- What happens if the mapping evolves?

Use an information integration tool

- Tools that provide mechanism to specify mapping using **SQL**
 - Data can be processed in multiple steps that may include running arbitrary code on it.
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Classifications

Data Replication and Timeliness

Data Replication and Timeliness

What is the result of the integration?

- **Materialization**
 - A representation of the result of the integration is loaded from the sources into another database
 - Such representation represents a snapshot of the data at a specific time.
- **Virtualization (Mediator-Based)**
 - No concrete representation of the information is loaded (data are not moved)
 - All tasks are performed **on-the-fly** over the data sources (always up-to-date)

Materialized Information Integration

In **materialized** data integration, a **representation** of the result of the integration process is loaded from the sources

Such loading process can be either lazy or continuous

- Continuous loading is often used in data warehousing.
- Lazy loading is used to perform specific tasks or to obtain snapshots of the data.

Often performed using specific tools

- Extraction, transformation, and loading (ETL) tools that allow one to specify what data to extract (declaratively) and how to do it (procedurally)

In the literature: **Data Exchange**

Virtual Information Integration

In **virtual** data integration, all tasks are performed directly at the sources.

- Users' tasks are specified over the **global schema** but data stays at the sources
 - No unnecessary movement of data.
 - No need to update a local representation of the integration process.
 - Does not provide a mechanism for snapshotting data.
- Multiple software components perform the tasks over the sources
 - **Wrappers** are used to access the sources and present data in a standard format, thus solving physical heterogeneity.
 - A **mediator** combines and reconciles information coming from the wrappers, thus solving conceptual heterogeneity.

Approaches to Information Integration

Approaches to Information Integration

The problem of integrating information from multiple sources is as old as business organizations.

Over the years, different technological approaches have been proposed.

- Some may still be relevant today, depending on the size of the organization.

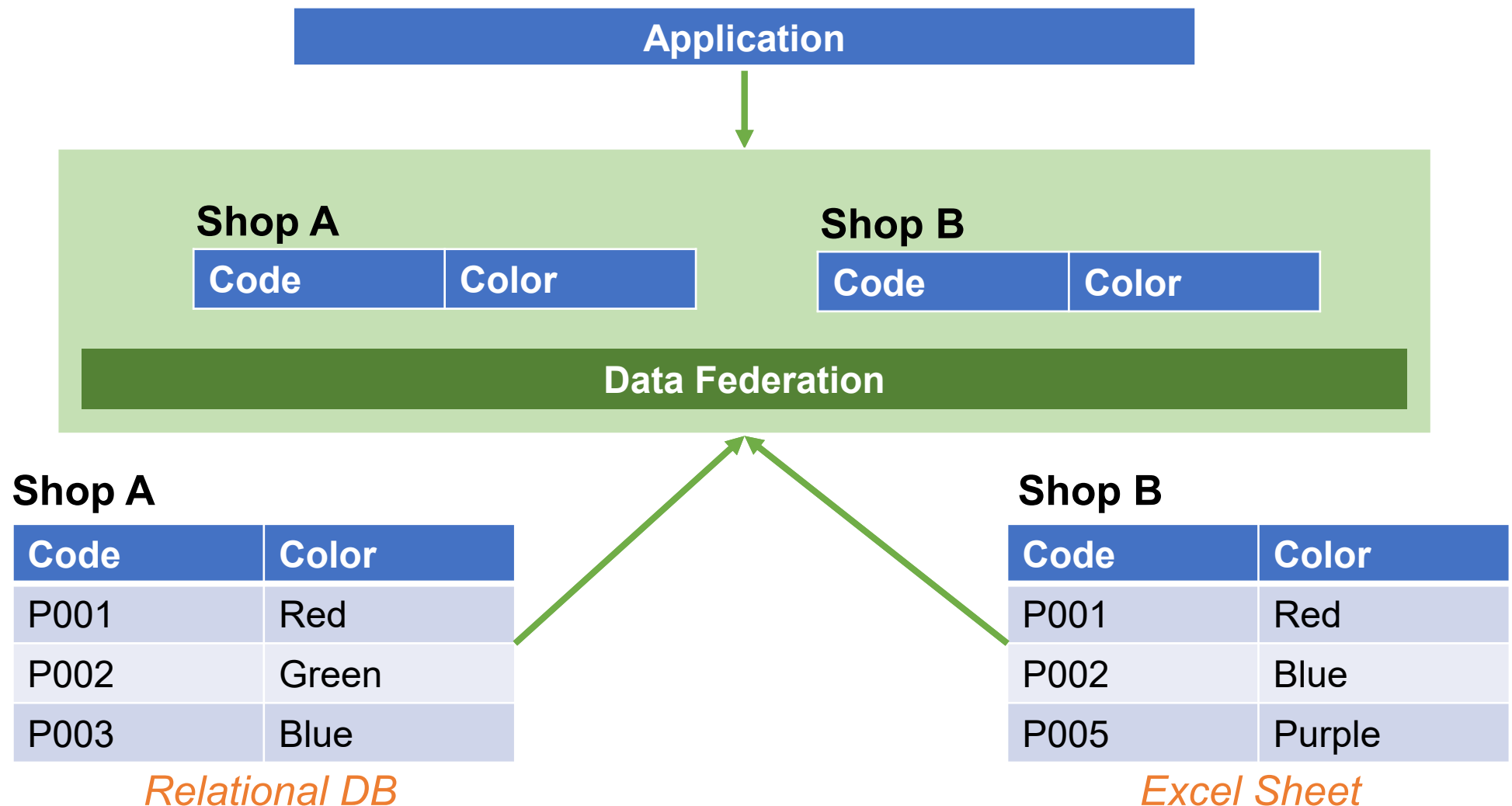
We survey some relevant approaches under the lens of the notions introduces in the previous part of this lecture.

Data Integration vs Data Federation

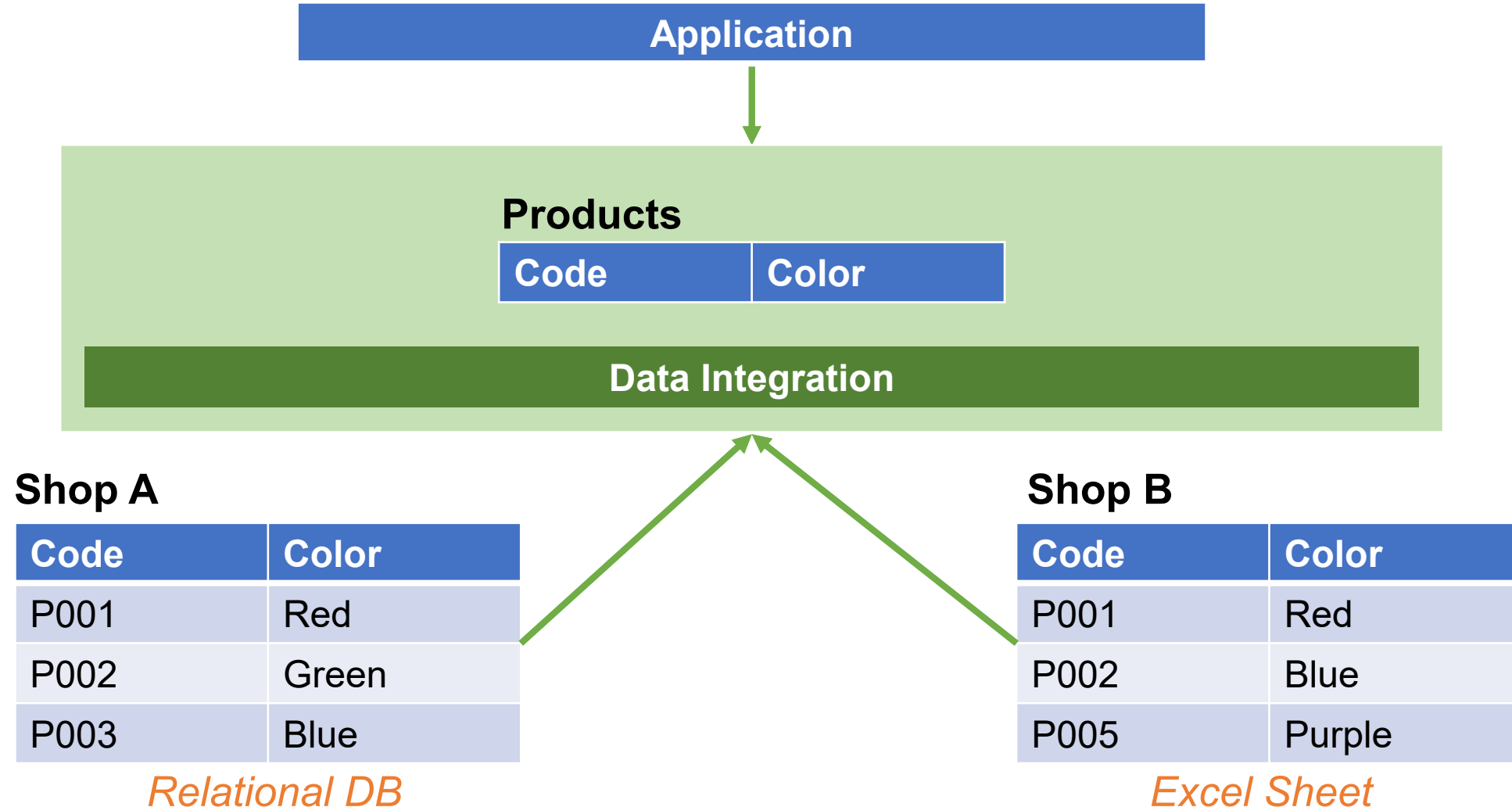
- A common source of confusion is the difference between **data integration** and **data federation** tools.
- **Data federation tools:** provide **access** to data stored in multiple sources.
- **Data integration tools:** **reconcile** data stored in multiple, autonomous, and heterogeneous sources.

The difference is Logical (or Conceptual) Transparency (the way data is stored at the sources does not affect the way the information is presented to the end user)

Data Federation



Data Integration



Achieving Logical Transparency

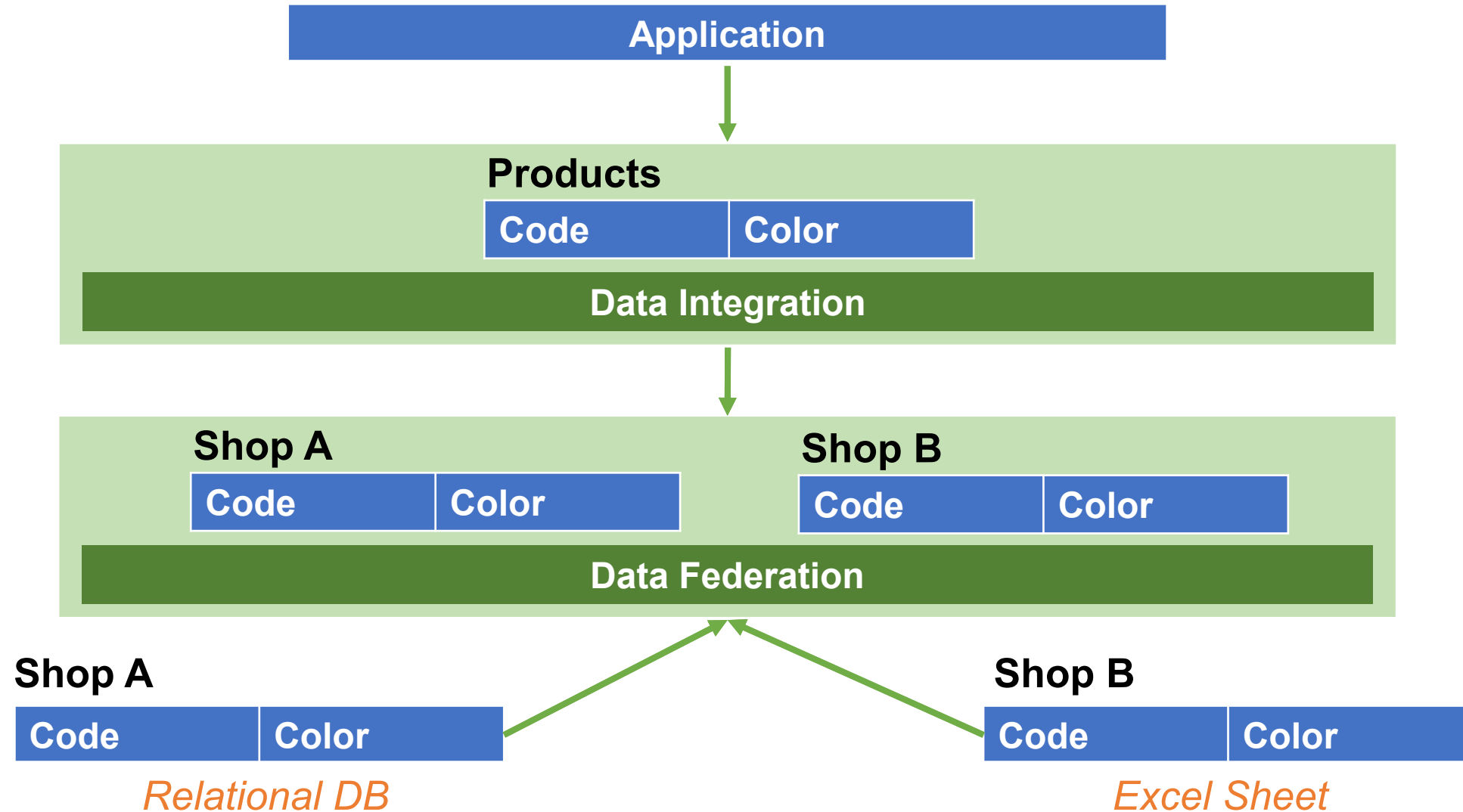
Data Integration tools achieve Logical Transparency using:

- A **Global Schema** providing a view of the scenario of interest independent from the way the **sources** store the data
- A **Mapping Layer** that reconciles the **source schema** with the **global schema**
- Rich formalisms for the specification of **global schemas** and **mappings**.

Data federation and data integration are often orthogonal.

- Data integration tools may be built on top of data federation tools.
- Often, we can use data federation tools to build the **wrappers** of an information integration tool.

Data Integration + Data Federation



Challenges of Information Integration

Challenges of Information Integration

Information integration may be a very challenging task both technologically and from a theoretical point of view

In the next slides, we briefly discuss some of the challenges of information integration

This list has no claim of completeness!

Indeed, Many challenges of information integration are domain specific.

Language-Related Challenges

- Expressive power
 - What formalisms should we use to express **global schema** and **mapping**?
 - Languages too rich may give rise to hard or undecidable tasks
 - Languages too simple may not be enough to define the desired properties.
- Modelling problem
 - Fixed a formalism, what is the best way to capture the intended properties?
- Automatic definitions
 - Can we devise tools to automatically define **mappings**?

Data-Management Challenges

- Efficient Query Processing
 - Depending on how conceptually different are the **sources**, information integration systems may need to specify very complex queries.
 - How can we **speed up** the computation of such queries?
 - How can we materialize the right amount of information?
- Information Management
 - How can we updates, deletions, and insertions expressed over the **global schema**?
 - What does it mean to update the **sources** w.r.t. the **global schema**?
- Data source wrapping
 - How can we efficiently define wrappers for data **sources**?
 - Can we do it automatically, i.e., can we automatically discover the right types?

Information Integration Challenges

- Virtual Data Integration
 - How do we translate a query over the **global schema** into a query for the **sources**?
- Materialized Data Integration
 - What kind of representation should we load from the **sources**?
- Global Schema Constraints
 - Can we define further requirements over the **global schema**, independently from the **mapping**?

Credits

The presentations of this module of the course are based on the original slides of Prof. Giuseppe De Giacomo, Prof. Diego Calvanese, Prof. Marco Console, Prof. Maurizio Lenzerini, and Prof. Domenico Lembo.
